

Brain gray matter changes across chronic pain trajectories: MRI evidence from 35,000 individuals

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Abstract

Chronic pain affects brain gray matter structure, but the magnitude, pattern, and reversibility of brain changes upon pain recovery remain unclear because of inconsistent findings from small-scale studies. We analyzed data from 35,052 UK Biobank participants who completed chronic pain assessments at 2 time points, approximately 9 years apart, evaluating pain lasting 3 months or more, across 8 body sites. Brain magnetic resonance imaging data were available from the second time point. Participants were grouped into 4 categories: Pain-free controls, New chronic pain, Persistent chronic pain, and Resolved chronic pain. We examined relationships between these categories and 313 brain-wide measures across cortical surface area, cortical thickness, and subcortical volume. New chronic pain was associated with surface area reductions in sensorimotor, temporo-parietal, and medial and lateral prefrontal cortices. Persistent chronic pain showed larger and more widespread area reductions and mixed changes in cortical thickness and subcortical volumes, including cerebellar atrophy and hypertrophy in the thalamus and basal ganglia. Resolved pain demonstrated reversibility of most brain changes, except sustained surface area reductions in the primary somatosensory cortex. No areas showed effects consistent with a predisposition to pain. Brain structural changes in chronic pain reflect adaptive neuroplasticity rather than irreversible damage or preexisting vulnerability. Most changes appear reversible with pain resolution, underscoring the importance of effective pain treatment to alleviate symptoms and support recovery of brain structure and function.

Keywords: Chronic pain, Brain structural changes, Pain resolution, Atrophy, Hypertrophy, Cortical thickness, Cortical surface area

1. Introduction

Chronic pain has been construed as causing irreversible damage to the brain.^{76,112} Long-term pain is associated with brain circuit alterations that promote long-term behavioral changes, including withdrawal, avoidance, and anhedonia.^{15,52,77,89,90,101,125} Animal models show structural brain changes, such as remodeling of motivational systems,^{18,65} with similar findings in human magnetic resonance imaging (MRI) scans of patients with pain.^{12,67} Nevertheless, the specific brain regions implicated and plasticity of these changes remain unclear.

Several studies have examined chronic pain-related associations with changes in regional brain gray matter,^{8,31,63} including

a meta-analysis that found reduced gray matter volume throughout the brain across pain conditions.¹²⁷ However, the constituent studies had small samples⁸⁸ and heterogeneous data. Prior meta-analyses have depended on summary-level data with low fidelity and anatomical precision,¹³⁹ and the largest meta-analysis detected no significant chronic pain effects in its primary analyses.⁶³ A recent well-powered study of 20,000 participants found widespread cortical surface area reductions and cortical thickness increases across multiple chronic pain conditions.¹⁷

An important unresolved question is whether gray matter changes precede or follow the onset of pain, and whether they are reversible when chronic pain resolves. Several studies in support of pain-predisposing neurological factors have focused on measures of functional connectivity^{12,135} and brain structure.^{6,54,61,69,113,121} By contrast, early reports described gray matter alterations with pain as damage,^{6,10,33,69,76,112} implying detrimental and irreversible changes. However, longitudinal studies have reported that pain-related brain structural changes were partially reversed with pain treatment.^{61,113,121} Thus, gray matter differences may be a reversible consequence of chronic pain, but this has yet to be assessed in a well-powered, population-based study.

We used the UK Biobank cohort to identify structural effects consistent with 4 types of pain–brain relationships based on pain trajectories across 2 time points, which we call “meta-hypotheses.” We then identified the most likely pattern for each region across the cortex and subcortex: (1) no effects; (2) predisposing risk for chronic pain; (3) irreversible changes with

Sponsorships or competing interests that may be relevant to content are disclosed at the end of this article.

K. Zorina-Lichtenwalter and C. I. Bango shared first-authorship.

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Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.painjournalonline.com).

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<http://dx.doi.org/10.1097/j.pain.0000000000004125>

chronic pain consistent with brain “damage”; and (4) reversible adaptations to pain.

2. Methods

2.1. Overview and hypotheses

To assess regional relationships between chronic pain and brain structure, we examined structural MRI-based changes in a large population sample (see Participants below). Imaging-derived phenotypes (IDPs) consisted of cortical volume, surface area, and cortical thickness for each of 148 parcels defined in the Destrieux atlas,³⁹ as well as 17 subcortical volume parcels, mapped using Freesurfer’s automatic segmentation tool, *aseg*⁵⁰ (eTables 2–5, <https://links.lww.com/PAIN/C582>). Chronic pain (defined as pain persisting >3 months) was assessed on 7 body sites, at 2 time points on average 9 years apart (range: 4–14 years), with MRI imaging at time 2 (Fig. 1). Subgroups of at least 5000 participants reported chronic pain at the baseline assessment (time 1), MRI assessment (time 2), or both. We grouped participants into 4 categories: (1) Control (no pain at either timepoint), (2) New chronic pain (pain at time 2 with no pain at time 1), (3) Persistent chronic pain (pain at both time points), and (4) Resolved chronic pain (pain at time 1 but not pain at time 2) (Fig. 1A). In addition to the 3 categories for chronic pain in any body site, we derived these categories for each body site (Table 1).

Inferences about the nature of gray matter–associated changes depend on both (1) identifying directional changes in those with pain and resolved pain using standard contrasts and (2) evaluating the relative likelihood of different functional hypotheses based on patterns of differences and equivalencies across pain groups. Since the study only included imaging at time 2, we used cross-sectional group differences to evaluate which patterns were most compatible with each meta-hypothesis (Figs. 1B and C). Different patterns of gray matter changes are differentially compatible with different meta-hypotheses (Fig. 1C). For example, regions with gray matter loss in Persistent vs Control groups but no differences between Resolved vs Control groups show a pattern consistent with reversible changes, and

inconsistent with conferring a predisposition for future pain. By contrast, regions with gray matter loss in Persistent vs Control groups and equivalent reductions in Resolved vs Control groups are consistent with irreversible gray matter loss or a predisposition to future pain. However, if these regions show little effect of New pain vs Controls, this effect is inconsistent with predisposition, as brain changes increase after longer-duration pain. Following this logic, we identified 15 different subhypotheses of the 4 meta-hypotheses described above, patterns of gray matter increases or decreases across the 4 groups, and we used Bayesian model comparisons to quantify statistical evidence for each, identifying brain regions in which the evidence strongly favored one meta-hypothesis over the others (see Section 2.8 below).

2.2. Participants and study design

Participants were drawn from the UK Biobank Cohort (data use application 43,851, <https://biobank.ndph.ox.ac.uk/>). Approximately 500,000 participants aged 40 to 69 were recruited from the population of the United Kingdom. All participants completed a baseline assessment between 2006 and 2010 during which demographic and health information as well as biological samples were collected.²⁴ A subgroup of ~40,000 participants returned for a second assessment between 2014 and 2022, during which neuroimaging data using MRI were collected. At both assessments, participants were also asked to complete a brief pain questionnaire, assessing the presence of acute and chronic pain (see below for details). In this study, we included 35,052 participants (mean age at baseline: 55.5 and at MRI assessment: 64.5 years [range: 45–82], 53.1% female, 97% White European, 0.6% Black, 0.79 South Asian, 1.2% Other/Mixed ancestry, and 0.29 East Asian) who had both neuroimaging and pain data, as described below and shown in Table 1.

2.3. Neuroimaging acquisition and data processing

Data acquisition and processing protocols for MRI are described in Reference 128. Briefly, participants underwent a 5-minute T1-

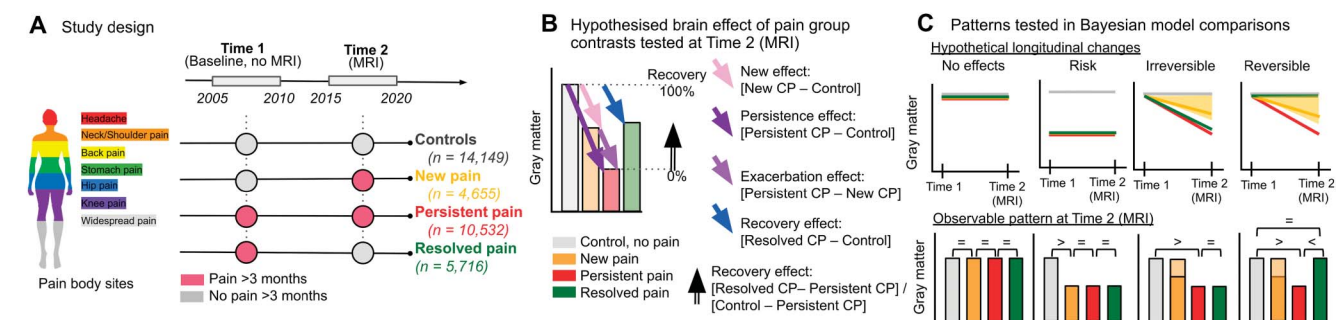


Figure 1. Study overview. (A) We compared gray matter measures across 4 groups, which were defined based on the pattern of chronic pain (CP) they exhibited across 2 time points that were on average 9 years apart. Pain-free Controls reported no chronic pain at either time point. The New chronic pain group reported chronic pain only at the second (MRI visit) session. The Persistent chronic pain group reported pain at both visits. The Resolved chronic pain group reported chronic pain at the initial visit, but no chronic pain at the second (MRI) visit. For all groups, chronic pain was defined as pain persisting for more than 3 months in any of 7 locations (head, neck, back, stomach, hip, knee, or widespread). The sample sizes (n) for each group are shown. (B) Bar plot: Hypothesized brain effects of different chronic pain group contrasts. New effect is New chronic pain – Control; Persistence effect is Persistent chronic pain – Control; Exacerbation effect is Persistent chronic pain – New chronic pain; Recovery effect is the ratio (Resolved chronic pain – Persistent chronic pain)/(Control – Persistent chronic pain). (C) Four types of effects (meta-hypotheses) defined by patterns of differences and equivalencies across the 4 groups. We used Bayesian model comparisons to compare evidence for each type of effect, regardless of direction (loss or gain of gray matter). For example, to be consistent with a predisposition to pain (the Risk panel), the gray matter values should be equivalent for New, Persistent, and Resolved groups and either lower or higher for these groups than Controls. For a Reversible effect, there must be a gray matter change in at least the Persistent group that is reversed when pain resolves, so that the Control and Resolved groups are equivalent. See Figure 3 for the complete set of 15 hypotheses. Mixtures of these hypotheses are possible, although Bayesian model comparisons identified regions for which 1 meta-hypothesis was substantially more likely than all others.

Table 1
Sample demographics.

	Total sample (n = 35,052)	Pain-free controls (n = 14,150)	Unresolved chronic pain (n = 10,532)	New chronic pain (n = 4656)	Resolved chronic pain (n = 5716)
	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)
Sex, female/male (%)	18,597/16,455 (53.1/46.9)	7044/7105 (49.8/50.2)	6088/4444 (57.8/42.2)	2399/2256 (51.5/48.5)	3066/2650 (53.6/46.4)
Overall age	64.5	64.8 (7.6)	63.9 (7.7)	64.9 (7.9)	64.5 (7.5)
Female age	63.8 (7.5)	64.1 (7.5)	63.2 (7.5)	64.1 (7.9)	63.8 (7.3)
Male age	65.3 (7.7)	65.4 (7.7)	64.9 (7.8)	65.7 (7.7)	65.3 (7.7)
Years elapsed	9.0 (1.8)	9.0 (1.8)	9.0 (1.8)	9.0 (1.8)	9.0 (1.8)
Ethnic background					
White	34,012	13,747	10,168	4542	5553
Black	194	78	72	20	24
South Asian	278	103	102	33	40
Other/Mixed	465	133	163	50	75
East Asian	103	44	27	10	22
Pain site (localized)					
Back pain	1733		336	595	802
Hip pain	650		58	404	188
Knee pain	2315		454	1050	811
Neck or shoulder pain	1590		215	625	750
Headache	1012		314	159	539
Stomach pain	272		27	107	138
Pain all over the body	194		97	50	47

weighted MPRAGE scan using a Siemens Skyra 3T scanner and standard Siemens 32-channel RF head coil (resolution 1x1x1mm, field-of-view 208 × 256 × 256) in one of 4 imaging assessment centers, and images were processed as described in the UK Biobank Brain Imaging Documentation (https://biobank.ctsu.ox.ac.uk/crystal/crystal/docs/brain_mri.pdf).

We obtained processed IDPs: quantitative measures of brain gray matter structure made available through the UK Biobank's automated image processing pipeline.³ The IDPs consisted of gray matter volume, surface area, and cortical thickness, each for 148 parcels defined in the Destrieux atlas,³⁹ as well as gray matter volume for 17 subcortical parcels, mapped using Freesurfer's automatic segmentation tool, aseg,⁵⁰ with quality control applied as described in Supplementary Materials, <https://links.lww.com/PAIN/C581>.

The full set of 461 regional measures is described in eTables 2 to 5, <https://links.lww.com/PAIN/C582>.

2.4. Definition of chronic pain groups

Participants completed a pain questionnaire at the baseline assessment and at the MRI assessment. They answered the question (data-field ID 6159): "In the last month have you experienced any of the following that interfered with your usual activities? (You can select more than one answer)." Possible answers were (1) Headache, (2) Facial Pain, (3) Neck or shoulder pain, (4) Back pain, (5) Stomach or abdominal pain, (6) Hip pain, (7) Knee pain, (8) Pain all over the body, (9) None of the above, and (10) Prefer not to answer. If participants selected one or more pain sites, they were asked if they had experienced the site-specific pain for more than 3 months (data-field IDs 3571, 4067, 2956, 3799, 3414, 3773, 3404, 3741). For each time point (baseline and MRI), we defined as cases those participants who reported having pain at any of the 8 sites for more than 3 months. We defined as controls those participants who selected "None of the above" for data-field ID 6159. We excluded participants if (1) They reported pain for greater than 1 month but fewer than 3 months

(subacute-stage pain) or (2) They responded "Do not know" or "Prefer not to answer" for either data-field ID 6159 or the follow-up site-specific questions. We further grouped cases using 2 time points: New chronic pain for cases only at time 2, Persistent chronic pain for cases across both time points, and Resolved chronic pain for cases only at time 1 (**Fig. 1A**).

2.5. Diagnosed pain conditions

To compare brain measures across diagnosed pain conditions, we included a list of 17 pain conditions from hospital records (ICD10 codes, UKB df. 42001) and the Experience of Pain questionnaire (UKB category 154). To ensure that we were capturing clinically established diagnoses, the latter were only included if they were given as answers to the question: "Have you ever been told by a doctor that you have had X condition?" Diagnosed pain conditions were assessed from clinical records and thus were not assessed specifically at time 2 functional magnetic resonance imaging, (fMRI), but including them enabled us to examine subsets of participants with both self-reported chronic pain and a specific clinical diagnosis to corroborate the fMRI scan-time pain measure. The included pain diagnoses are listed in Table S23, <https://links.lww.com/PAIN/C582>, with osteoarthritis, migraine, and stomach pain being the most prevalent. For conditions available from both hospital records and the questionnaire, data were combined such that if it was present in one of these records for a given individual, that individual was considered a case for that condition (with the number of unique participants with each case captured in the Total Sample Size column).

2.6. Estimating regional gray matter in pain groups

We regressed each IDP on dummy-coded regressors for each of the 3 pain groups, with Controls as the reference, adjusting for sex, age, age-squared, intracranial volume, testing center, and table position. In these regression models, the intercept

estimated the mean of the Control group, and group regressors estimated the means of each pain group compared with Controls. To facilitate reporting of standardized effect sizes, gray matter outcomes were z-scored.

We performed separate regression models on each of 17 subcortical gray matter volume parcels and 148 cortical parcels mapped to the Destrieux atlas for each of 3 cortical measures: surface area, cortical thickness, and cortical volume (461 regressions total), using a mapwise false discovery rate (FDR) correction thresholded at $q < 0.05$. As cortical volumes are a derived measure, ie, the product of cortical area and thickness, we only used them as part of a follow-up analysis to investigate the possibility of compensatory plasticity between cortical surface area and cortical thickness.

Regression models used the general linear model (“glm”) function in R’s “stats” package, version 3.6.2. Dependent variables (IDPs) were standardized (z-scored). To account for multiple tests, a FDR adjustment was applied for all tests within each type of gray matter measure (surface area,¹⁴⁷ cortical thickness,¹⁴⁷ and subcortical volume¹⁷) and thresholded at $q < 0.05$. In secondary analyses, we assessed whether age, sex, and head movement contributed to patterns of pain-related gray matter changes.

To assess the relevance of the estimates from these models to diagnosed pain conditions, in additional analyses we focused on a subsample of chronic pain sufferers with at least one pain diagnosis. For these analyses, we defined controls as participants with neither pain reported in any body site nor any pain diagnosis. Then, we reran the regression models, one for each region of interest (ROI) in each modality (cortical surface area, cortical thickness, and subcortical volumes), and we examined correlations between effect sizes in the full dataset and the subset with a confirmed clinical diagnosis.

2.7. Pain group comparisons

To estimate and visualize the mean levels and standard errors for all groups (pain-free Controls, New, Persistent, and Resolved Chronic Pain), we ran a variant of the regression model with binary indicator variables for each of the 4 groups, mean-centered covariates, and no intercept, thereby obtaining covariate-adjusted group means. For this analysis, to estimate gray matter levels in their original units, we did not standardize the gray matter outcomes. To estimate the proportion of resolution of each brain effect, we calculated the ratio of differences as follows: (Resolved – Persistent)/(Controls – Persistent). Here, a score of 0 indicates no resolution and a score of 1 complete (100%) resolution. Negative scores are possible because of both statistical error and neotrophy (eg, increases with Resolved vs Control in the absence of Persistent vs Control differences). We only calculated these proportions for IDPs with significant Persistent pain effects estimated in the main model (Estimating regional gray matter).

Given previous reports of different gray matter patterns for different body sites and different pain conditions,^{17,60} we looked for these differences using analysis of variance (ANOVA) tests. We estimated covariate-adjusted mean ROI values for chronic pain in each body site (exclusive of others), using pain reports at the imaging time point. In addition, we estimated covariate-adjusted mean ROI values for each clinical pain diagnosis (exclusive of others). Within each analysis—one for differences across body sites and one for differences across pain diagnoses—we ran an ANOVA omnibus test for each ROI. For body sites, this was a model comparison with a difference of 7 degrees of freedom. In

the full model, each ROI is regressed on covariates and 7 dummy-coded individual predictor variables spanning the space of all possible differences across 8 body sites. In the reduced model, the ROI is regressed on covariates only. In addition, we examined pain group-by-body site interactions, also using a seven-degree-of-freedom test to compare the full model with interactions to the reduced model with main effects of the pain group, body site, and covariates. We then compared the ANOVA R^2 between the full and reduced models using an F-test and thresholded the results for multiple comparisons across regions at FDR $q < 0.05$. If the omnibus test showed a significant difference in R-squared in any ROI, we ran a post hoc one-degree-of-freedom test for that ROI, specifying the full model with one pain condition at-a-time as the predictor and the reduced model as above. The post hoc test showed whether the mean of a given pain site was different from the average mean across the other sites. For pain diagnoses, we performed the same type of model comparison, using a full model that included 16 predictors spanning the space of all differences across 17 diagnosis categories for the omnibus test. We additionally grouped pain diagnoses into 3 mechanistic categories: nociceptive, neuropathic, and nociplastic.⁷⁵ Although patients with a given diagnosis may have mixed features of all 3 types, our categorization was based on increased prevalence/likelihood of 1 of the 3 with certain diagnoses: nociceptive (osteoarthritis, knee arthritis, hip arthritis, and rheumatoid arthritis), nociplastic (fibromyalgia), and neuropathic (neuralgia and nerve compression pain), as most recently established by the International Association for the Study of Pain.

2.8. Bayes factor ratios

To confirm that regions showing no significant effect for the Resolved-versus-Controls regressor had no difference between these 2 groups, we calculated the following Bayes Factors: one for a null (Resolved-versus-Controls difference is zero), and one for an alternative model (Resolved-versus-Controls difference is not zero). Then, we divided the former by the latter to obtain a ratio. We set the following ranges: weak evidence in favor of the alternative, 0.33 to 1; moderate evidence in favor of the alternative, 0.1 to 0.33; strong evidence in favor of the alternative, 0 to 0.11; weak evidence in favor of the null, 1 to 3; moderate evidence in favor of the null, 3 to 10; strong evidence in favor of the null, >10 .

2.9. Bayesian posterior estimation using Bayes factors

To compare meta-hypotheses, we first specified 15 distinct patterns of gray matter changes across the 4 groups (“models” or “subhypotheses”) using categorical contrast variables, so that we could quantify the likelihood of each model and compare them (Table S1, <https://links.lww.com/PAIN/C582>). Null effects are consistent with a pattern in which all groups have equal gray matter values. Predisposing risk factors are consistent with a difference between Controls and Pain groups (reductions or increases), but no further differences as a function of pain persistence or resolution. Irreversible changes involve differences between Controls and pain groups, but no changes between Persistent and Resolved pain, with changes occurring either with New pain or only with Persistent pain (2 patterns). Reversible changes involve 8 distinct patterns, characterized by pain-related changes that occur earlier (with New pain) or later (only with Persistent pain), are progressive (eg, Control $>$ New $>$ Persistent) or not, and are fully or partially reversible. Finally, we specified other possible patterns as “complex changes,” such as changes

in New and Resolved groups but not Persistent. We did not observe any regions with evidence for this last group of hypotheses. We evaluated evidence for each meta-hypothesis using Bayesian model comparisons,¹⁰⁹ ie, the likelihood of each meta-hypothesis given the pattern of gray matter density across the 4 participant groups (Controls, New chronic pain, Persistent chronic pain, and Resolved chronic pain; Supplementary Materials, <https://links.lww.com/PAIN/C581>).

2.10. Degree of reversibility

To characterize the degree of reversibility in gray matter loss for regions with significant Persistent chronic pain effects, we computed the ratios of differences in gray matter between covariate-adjusted means for (Resolved – Persistent chronic pain)/(Controls – Persistent chronic pain), for IDPs with significant persistent effects in the main model.

3. Results

3.1. Regional gray matter estimates in pain groups

3.1.1. Cortical surface area

The New chronic pain vs pain-free Controls contrast showed 26 areas with significantly decreased surface area after adjusting for covariates, including the left primary motor cortex (M1), bilateral primary and right secondary somatosensory cortices (S1 and S2), right middle frontal gyrus covering parts of the dorsolateral prefrontal cortex (dlPFC), bilateral anterior and posterior midcingulate cortices (a/pMCC), bilateral inferior parietal lobule (IPL), “default-mode” network regions (pregenual anterior and posterior cingulate cortices, ACC [part of ventromedial prefrontal cortex, vmPFC] and PCC), temporoparietal junction (TPJ), and large portions of occipital and lateral cortices (**Fig. 2A**, Table S2,

<https://links.lww.com/PAIN/C582>), with the largest effect size in the right S1 (postcentral gyrus).

Persistent chronic pain vs Controls showed significant decreases in surface area in 88 cortical regions. These regions included the areas showing significant decreases in all New chronic pain vs Controls reported above, with additional decreases in the dlPFC and ventromedial PFC (vmPFC), bilateral insula, anterior MCC (aMCC), bilateral orbitofrontal cortex (OFC), and bilateral superior parietal lobule (SPL) (**Fig. 2B** and Table S2, <https://links.lww.com/PAIN/C582>). Resolved chronic pain vs Controls showed surface-area reductions only in right S1 (right postcentral gyrus and sulcus, **Fig. 2C** and Table S2, <https://links.lww.com/PAIN/C582>).

Of 148 IDPs, 141 showed moderate-to-strong evidence in favor of the null hypothesis Bayes Factor Ratio (BFR), range 3–58, Fig. S1A, <https://links.lww.com/PAIN/C581>, indicating a lack of gray matter differences between Resolved chronic pain and Controls. Convergent, of IDPs with significant Persistent chronic pain effects in the main model (**Fig. 2D** and Fig. S2B, <https://links.lww.com/PAIN/C581>), many show reversibility ratios close to 1, most consistent with complete resolution. Approximately 24% of regions showed greater than 90% resolution, 40% of regions showed between 50% and 90% resolution, and 36% of regions showed less than 50% resolution, although many of these still showed nonsignificant differences between Controls and Resolved chronic pain. Taken together with the facts that (1) only S1 showed significant reductions in Resolved chronic pain vs Controls and (2) the majority of BFRs supported the hypothesis of no difference between the Resolved and Control groups, these results suggest that most pain effects on surface area are reversed when pain resolves. Analyses of pain trajectories limited to those with pain on each individual body site (one analysis per body site) had minimal detectable effects, with most not passing the FDR $q < 0.05$ threshold (Fig. S22, <https://links.lww.com/PAIN/C581>).

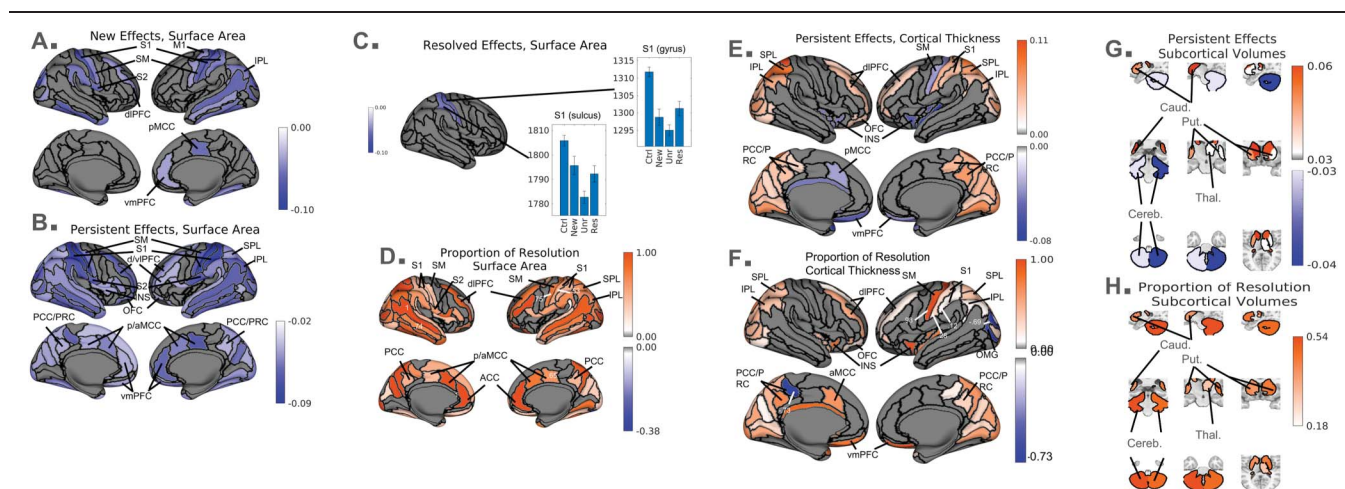


Figure 2. Gray matter changes with chronic pain. (A–C) Comparisons between Controls and each chronic pain (chronic pain) group in cortical surface area. (A) Control – New chronic pain. (B) Control – Persistent chronic pain. (C) Controls – Resolved chronic pain. Blue colors indicate significant gray matter reductions with pain, at FDR 5%. The bar plots show average surface area values across the 4 groups for 2 regions of the primary somatosensory cortex (S1), with ~50% resolution of surface area reductions with Resolved as compared with Persistent chronic pain. (D) The proportion of resolution of surface area reductions, defined as (Resolved – Persistent)/(Controls – Persistent). 0 indicates no resolution, 1 indicates complete resolution, values in-between indicate partial resolution, and a negative ratio indicates that the Resolved group has a larger deviation from Controls than does the Persistent group from Controls (in the right posterior transverse collateral sulcus, not shown, –0.377). The remaining panels show increases (orange, hypertrophy) and decreases (blue, atrophy) in cortical thickness (E and F) and subcortical volumes (G and H). (E) Shows significant changes and their magnitude, and (F) shows proportion of resolution. Both show Control – Persistent chronic pain differences only, as no other contrasts (Control – New and Control – Resolved) showed significant differences. In (F), the negative ratio in blue for the OMG and PRC here also indicates that [Resolved – Control] are larger than [Persistent – Control] effects, signaling possible neurotrophism with pain resolution. ACC, anterior cingulate cortex; a/pMCC, anterior/posterior midcingulate cortex; Caud., caudate; Cereb., cerebellar cortex; dlPFC, dorso/ventrolateral prefrontal cortex; FDR, false discovery rate; INS, insular cortex; IPL, inferior parietal lobule; M1, primary motor cortex; OFC, orbitofrontal cortex; OMG, occipital middle gyrus; PCC/PRC, posterior cingulate cortex/precuneus; Put., putamen; S1/2, primary/secondary somatosensory cortex; SM, sensory-motor cortex; SPL, superior parietal lobule; Thal., thalamus; vmPFC, ventromedial prefrontal cortex.

In analyses restricted to chronic pain participants with confirmed pain diagnoses ($N = 12,949$), the brain effect sizes were slightly smaller than in the full sample, but the same pattern of changes was found for Persistent vs Control (Fig. S21, <https://links.lww.com/PAIN/C581>). Furthermore, the pattern of effect sizes across brain regions was highly correlated with the pattern in the full analysis: $r = 0.95$ for Persistent pain, $r = 0.84$ for New pain, $r = 0.83$ for Resolved pain.

In comparisons of surface area differences in gray matter across pain body sites (Fig. S10, <https://links.lww.com/PAIN/C581>) and pain diagnoses (Fig. S14, <https://links.lww.com/PAIN/C581>), several brain regions with significant effects in the main analysis showed reduction in all pain groups compared to pain-free controls. ANOVA model comparisons, both for main effects and interactions, showed no significant differences across pain body sites or diagnoses in these regions, with the largest ANOVA R^2 difference for both sets being 0.003. This indicates that a negligible amount of variance in gray matter is explained by differences across body sites. Analyses across putative nociceptive, neuropathic, and nociplastic categories (Fig. S17, <https://links.lww.com/PAIN/C581>) also showed no differences (FDR-corrected $P > 0.8$ for all regions).

3.1.2. Cortical thickness

We observed no significant differences in cortical thickness for New chronic pain or Resolved chronic pain vs Controls. However, Persistent chronic pain vs Controls identified 59 significant regions, with a mix of decreased and increased cortical thickness (Fig. 2E and Table S3, <https://links.lww.com/PAIN/C582>). Seventeen regions showed decreases, including the bilateral insula, aMCC, vmPFC, and left M1. Forty-two regions showed increases, including the left S1, bilateral dLPFC, PCC, and bilateral IPL and SPL. The largest decreases were in the bilateral anterior insula, with the largest increases in the parietal cortex (bilateral superior lobule, SPL, and right intraparietal sulcus, IPS).

Of 148 IDPs, 126 showed moderate-strong evidence in favor of the null hypothesis for the Resolved chronic pain vs Control contrast (BFR, range 3–26.5, Fig. S1B, <https://links.lww.com/PAIN/C581>), consistent with resolution of gray matter changes with pain resolution. Reversibility ratios revealed that the most clearly reversible changes occurred in M1, vmPFC, and PCC (>50% reversible), while the least reversible were in the parietal and occipital regions (<50% reversible; Fig. 2F; Supplementary eFig. 2C, <https://links.lww.com/PAIN/C581>, Table S3, <https://links.lww.com/PAIN/C582>). The occipital middle gyrus and precuneus (part of the PCC) were the only areas with negative ratios, indicating increased cortical thickness with both New and Persistent chronic pain and further gain with Resolved chronic pain, indicating possible new adaptations with pain resolution. Similarly to surface area, analyses in cortical thickness of trajectories grouped by individual body site had very few significant effects (Fig. S22, <https://links.lww.com/PAIN/C581>).

In analyses considering only individuals with confirmed pain diagnoses, the brain effect sizes were slightly smaller, but the correlations across brain regions with the effect sizes from the full set were 0.98 for Persistent pain, 0.89 for New pain, and 0.81 for Resolved pain (Fig. S21H, <https://links.lww.com/PAIN/C581>).

Comparisons of gray matter variations across pain body sites in ROIs with significant effects in the main analysis, both for main effects and interactions, did not show significant differences in the ANOVA omnibus test (Fig. S11, <https://links.lww.com/PAIN/C581>).

C581). However, 6 other ROIs, including the insula, did have significant differences. The R^2 for significant ROIs range from 0.003 to 0.006. Post hoc model comparisons showed significant increases for headache and decreases for CWP (Fig. S12, <https://links.lww.com/PAIN/C581>). Analyses of cortical thickness changes across diagnoses (Fig. S15, <https://links.lww.com/PAIN/C581>) revealed no significant regions. The largest R^2 for the effect of disorder type across brain regions $R^2 = 0.003$. Analyses across putative nociceptive, neuropathic, and nociplastic categories also showed no significant differences (Fig. S18, <https://links.lww.com/PAIN/C581>, FDR-corrected $P > 0.8$ for all regions).

3.1.3. Cortical and subcortical volumes

Cortical volume measures are derivative, cortical volume = surface area $\times$ cortical thickness, and their effects largely followed the surface area effect pattern (eTables 5 and 9, <https://links.lww.com/PAIN/C581>, Supplementary Materials, <https://links.lww.com/PAIN/C582>). In cortical volume, 7 regions showed Persistent chronic pain vs Control differences. These include increases in the right thalamus and bilateral striatum (putamen and caudate) regions and decreases in bilateral cerebellum (Fig. 2G and Table S4, <https://links.lww.com/PAIN/C582>). All 4 regions showed incomplete reversal of brain effects, with reversibility ratios of 0.178 to 0.538. Of 17 IDPs, 14 showed moderate to strong evidence in favor of the null hypothesis (no difference between Resolved chronic pain and Controls, BFR).

After subsetting by presence of at least one clinical pain diagnosis, the brain effect sizes were slightly smaller, but the correlations across brain regions with the effect sizes from the full set were $r = 0.98$ for Persistent pain, $r = 0.91$ for New pain, and $r = 0.87$ for Resolved pain (Fig. S21, <https://links.lww.com/PAIN/C581>).

In subcortical ROIs with significant effects for pain trajectories, comparisons across body sites (Fig. S13, <https://links.lww.com/PAIN/C581>) and pain diagnoses (Fig. S16, <https://links.lww.com/PAIN/C581>) did not have significant differences, both for main effects and interactions, in the omnibus ANOVA tests, with the largest R^2 being 0.002. Estimates trended numerically towards reductions in gray matter. Analyses across putative nociceptive, neuropathic, and nociplastic categories also did not show significant differences, including in the previously reported hippocampus,⁶⁰ whose uncorrected ANOVA omnibus P -values for left and right hemispheres are 0.16 and 0.24, respectively (Fig. S19, <https://links.lww.com/PAIN/C581>).

3.2. Bayesian model comparisons

We used Bayesian model comparisons evaluating the 4 meta-hypotheses (Fig. 3) and their constituent 15 hypotheses (Fig. 3A, Fig. S2A, <https://links.lww.com/PAIN/C581>).

3.2.1. Cortical surface area

Most regions with surface-area reductions in pain groups fit the pattern of reversible atrophy (meta-hypothesis 2, green in Figs. 3B–D), with at least 3:1 likelihood of reversible atrophy compared to the sum of all other meta-hypotheses. These regions include sensorimotor areas (left SM cortex and bilateral S2/operculum); medial areas associated with “cingulo-opercular” (aMCC and mid-insula) and “default mode” networks (pregenual ACC [part of vmPFC] and PCC); portions of the right dLPFC; and large areas of TPJ and lateral temporal cortices. Different brain areas

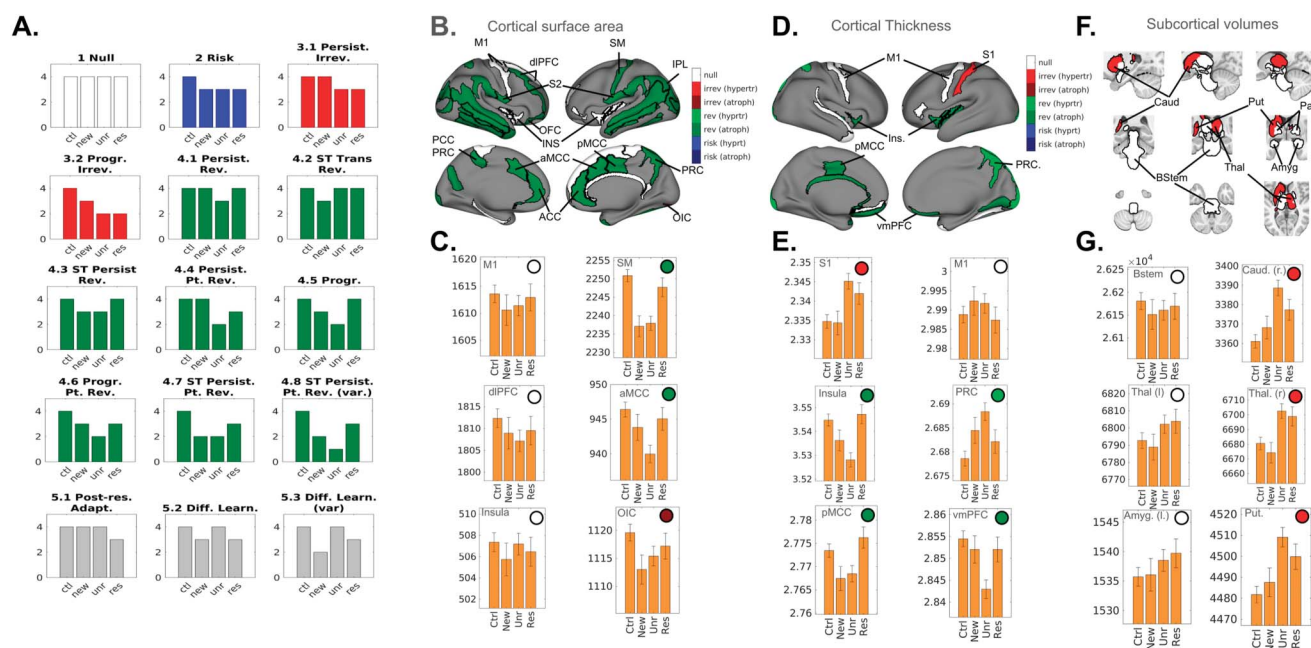


Figure 3. Bayesian Hypotheses. (A) Fifteen hypotheses about brain changes in chronic pain groups (Table S1, <https://links.lww.com/PAIN/C582>): 1. Null; 2. Risk; 3.1 Persistent Irreversible; 3.2 Progressive Irreversible; 4.1 Persistent Reversible; 4.2 Short-Term Transient Reversible; 4.3 Short-Term Persistent Reversible; 4.4 Persistent Partially Reversible; 4.5 Progressive; 4.6 Progressive Partially Reversible; 4.7 Short-Term Persistent Partially Reversible; 4.8 Short-Term Persistent Partially Reversible (variant); 5.1 Post-resolution Adaptation; 5.2 Differential Learning; 5.3 Differential Learning, variant. Bar colors and region colors in brain maps for cortical surface area, cortical thickness, and subcortical volumes (B, D, and F, respectively) correspond to the meta-hypothesis with the highest likelihood: white is “No effects”; blue is “Risk”; red is “Irreversible”; and green is “Reversible.” In (A), gray represents 1 of the 3 other possible hypotheses (5.1–5.3). In the brain maps, gray means none of the 4 meta-hypotheses reached the threshold of likelihood ratio, 0.75. In panels (C, E, and G), selected cortical and subcortical regions are shown with pain group average effects, adjusted for covariates. ACC, anterior cingulate cortex; a/pMCC, anterior/posterior midcingulate cortex; Caud., caudate; Cereb., cerebellar cortex; d/vIPFC, dorso/ventrolateral prefrontal cortex; INS, insular cortex; IPL, inferior parietal lobe; M1, primary motor cortex; OFC, orbitofrontal cortex; OIC, occipital inferior gyrus/sulcus; PCC, posterior cingulate cortex; PRC, precuneus; Put., putamen; S1, primary somatosensory cortex; SM, sensory-motor cortex; SPL, superior parietal lobe; Thal., thalamus; vmPFC, ventromedial prefrontal cortex.

show different patterns of reductions. DLPFC shows a pattern of progressive reduction (Control > New > Persistent) and recovery to Control- or near-Control levels (Figs. 3B and C). Somatomotor cortex decreases with New chronic pain and remains stable with Persistent chronic pain, whereas vmPFC shows reductions only with Persistent pain, although both reverse with Resolved pain. Only one region was most likely to show irreversible atrophy, the right occipital inferior gyrus/sulcus (OIC, Figs. 3B and C).

Several regions showed evidence in favor of null effects, ie, no differences across all groups, including right M1, superior frontal gyrus, medial SM, and much of the anterior and mid-insula (white in Figs. 3B and C). Areas shown in gray do not clearly favor one meta-hypothesis and thus do not permit strong inferences about their relationship with pain recovery and/or predisposition, although many show significant changes with pain in individual comparisons as reported above (see Supplementary Materials for discussion and individual hypothesis results, and Fig. S20, <https://links.lww.com/PAIN/C581> with ROIs endorsing different meta-hypotheses with stronger evidence, $BF > 10$).

3.2.2. Cortical thickness

Model comparisons showed > 3:1 evidence for reversible atrophy in bilateral insula, the pMCC, and vmPFC (Figs. 3D and E). Several occipital regions showed reversible hypertrophy, including the left precuneus. The only area to show irreversible hypertrophy was left S1. Several areas showed evidence for null effects, including bilateral M1, and Figure S17, <https://links.lww.com/PAIN/C581> with ROIs endorsing different meta-hypotheses with stronger evidence, $BF > 10$.

3.2.3. Subcortical volumes

Model comparisons showed evidence for irreversible hypertrophy in the right thalamus and caudate (Figs. 3F and G). They showed evidence for null effects in bilateral amygdala, left thalamus, right pallidum, and brainstem.

3.2.4. Relationships with age, sex, body mass index, signal-to-noise ratio, and head movement

To contextualize pain-related brain effects, we compared regional gray matter changes with those linked to head motion (Fig. S6, <https://links.lww.com/PAIN/C581>), age and sex (Fig. S7, <https://links.lww.com/PAIN/C581>), body mass index (BMI) (Fig. S8, <https://links.lww.com/PAIN/C581>), and inverted signal-to-noise ratio (SNR) (assessed by the region-level coefficient of variation in the T1, defined as the reciprocal of median [across brain voxels] of voxelwise mean intensity divided by voxelwise time-series standard deviation, UKB data field 25,743, Fig. S9, <https://links.lww.com/PAIN/C581>). The logic was that if pain effects were related to any one of these variables, then (1) we would observe large gray matter variations as a function of that variable, and (2) the regions with the strongest pain effects would be those with the strongest effects of that variable (a spatial correlation across brain regions).

Head motion was weakly associated with reduced surface area and strongly associated with reduced cortical thickness overall, which was the opposite direction as pain effects for cortical thickness. Moreover, the spatial pattern of effects showed weak ($r < 0.10$) correlations with pain effects in surface area and modest correlations ($r = 0.24$ – 0.27) with pain effects

in cortical thickness (Fig. S6, <https://links.lww.com/PAIN/C581>). Overall, this pattern of findings shows that head movement did not substantially confound pain-related effects on gray matter (see also eTables 10–18, <https://links.lww.com/PAIN/C582>).

Age was associated with reduced surface area and cortical thickness across brain regions overall, but pain effects showed weak spatial similarity to age-related changes, suggesting limited support for an “accelerated brain aging” model (Fig. S7, <https://links.lww.com/PAIN/C581>). On average, Persistent pain effects on surface area were roughly equivalent to 4 years of aging and slightly larger than age effects in cortical thickness. It is noteworthy, however, that the average age in our sample was 55 at the first time point and 64 at the second time point, which is somewhat older than many other neuroimaging studies of chronic pain.

Sex effects, while consistent with known patterns (larger surface area in males, thicker cortex in females), were largely uncorrelated with pain, except for a moderate “female-like” pattern in Persistent chronic pain.

On average, BMI was associated with reduced surface area, the same direction as pain effects, but it was not significant in cortical thickness. The pattern of BMI effects across brain regions was not correlated with either pain group’s effect in surface area, suggesting it is unlikely to explain the pattern of pain-related findings we observed. However, it was moderately correlated in cortical thickness, $r = 0.21$ with New pain and $r = 0.45$ with Persistent pain, raising the possibility that it explains some of the cortical thickness-related findings. Including BMI into the model as a predictor attenuated pain-related findings in surface area and rendered more ROIs significantly associated with Persistent pain in cortical thickness (eTables 19–20, <https://links.lww.com/PAIN/C582>). However, we believe that controlling for BMI in the main analyses is not appropriate, as it may reflect both a common cause of pain and cortical atrophy (reducing precision and power) and a common effect of pain and biological causes of gray matter reduction (eg, aging), which would induce bias in gray matter-to-pain relationships when controlling for it.³⁴

Low SNR (high coefficient of variation) was associated with lower surface area and cortical thickness values, as expected. These effects thus went in the opposite direction from pain effects for cortical thickness. Crucially, SNR was not correlated with pain effects (all r s < 0.1), and it is thus unlikely to explain pain findings. In addition, ROI-wise models showed that SNR had a moderate effect on pain effects in surface area and rendered 9 more ROIs significant for Persistent pain in cortical thickness. The head motion and SNR findings are consistent with the observation that participants in pain move more while in the scanner, but the pattern of pain-related findings cannot be explained away by either. See Supplementary Materials, <https://links.lww.com/PAIN/C582> for details.

4. Discussion

In a study of 35,052 individuals, we show that most structural brain changes are partially reversible consequences of pain. Reductions in surface area were notable in New chronic pain and more widespread in Persistent chronic pain. Changes in cortical thickness and subcortical volume included both partially reversible increases and decreases. These findings align with studies of neuroplasticity in nonhuman animals^{20,25,26,28,48,64,99,104,133} and may reflect adaptations to chronic pain¹²² that could influence nociceptive sensitivity,^{19,57,104} descending facilitation of

nociception,^{21,146,147} hypervigilance, threat avoidance, and reduced risk-taking.^{1,81} The brain effects we report are on the scale of $\beta_{std} \leq 0.1$ and required thousands of participants.⁸⁸ While these effects are unlikely to serve as person-level biomarkers, they continue to advance our understanding; small effects can have important implications for pain management and treatment. In particular, our results underscore that effective treatment may reverse pain-related brain gray matter changes.

In surface area, we observed chronic pain-associated losses in sensorimotor regions (S1, S1/M1, M1, S2), which have causal interactions with nociceptive systems and are targets for human pain neuromodulation,^{72,140} as well as areas like the dlPFC, vmPFC, MCC, TPJ, and IPL, which are involved in higher-order pain processing and complex cognition.^{5,53,62,74,84,95,138} Structural pain-related effects in these regions have been reported.^{17,51,73,80,86,96,117,121,142}

The Persistent pain group showed additional losses in frontal, occipital, temporal, and parietal regions, including the vlPFC, OFC, and insula, consistent with alterations noted in previous studies.^{14,22,54,62,111,117,138} Affected regions overlap with resting-state DMN,^{4,11} “salience,”^{45,87} and “visual” networks.¹²³ Together with the Bayesian model comparison results, the absence of effects in all but S1 of the Resolved group suggests reversal of chronic pain effects on surface area.

Cortical thickness changes are limited to Persistent chronic pain, suggesting these alterations take longer to develop than in surface area. Atrophic regions generally returned to control levels upon pain resolution, whereas hypertrophic regions, such as left S1, showed minimal recovery. Some hypertrophy occurred in areas with surface area loss (eg, parietal, occipital cortices, dlPFC, left S1). Among these, only dlPFC and S1 are traditionally associated with pain, showing both increased^{36,68,73,96,142,17} and decreased^{86,121} thickness. Regions with atrophy in thickness and surface area (eg, vmPFC, mACC, S1/M1) align with prior findings of gray matter loss. Insular thinning also corroborates earlier reports.^{46,51,121,23} Compared to Bhatt et al.,¹⁷ we observed both increases and decreases across more regions. Their “any chronic pain” group, most comparable to our Persistent chronic pain (Fig. 2B, E; Bhatt Figs. 1A and B), may have averaged effects that diverge between our New and Persistent chronic pain.

Our findings of divergent effects on surface area and cortical thickness (Fig. S5, <https://links.lww.com/PAIN/C581>) align with prior work¹⁷ and reflect their distinct genetic, phenotypic, and developmental bases.^{56,79,105,110,141} Changes in surface area precede cortical thickness, suggesting compensatory plasticity,^{78,106,129} in which surface contraction may trigger vertical expansion via reorganization¹⁴⁴ or reduced dendritic arborization.⁴⁰ Neuroinflammation, associated with cortical thinning and thickening,^{6,16,59,83,94,97} may explain spreading of surface area effects from the New to Persistent group, driven by multisite pain. The pattern of opposing cortical thickness and surface area changes, particularly in sulci, could suggest local adaptations or changes in myelination.¹⁰⁰ Our results may also be influenced by specific pain conditions like migraine, which has shown decreases in surface area and increases in cortical thickness.^{17,36,55,68}

S1 showed rare recovery-resistant effects, with interhemispheric differences. The right S1 exhibited persistent reductions in surface area but not in cortical thickness. Left S1 showed both weakly reversible surface area loss and almost irreversible cortical thickening. This divergence highlights hemispheric laterality in pain processing, shown previously for the thalamus and dlPFC,^{35,115,130} consistent with reported discrepancies for this

sensory brain region.¹³⁷ Some studies report no gray matter loss in S1,^{113,114} while others show decreases^{103,116,136} or irreversible thinning.^{44,121} Conversely, S1 has shown gain in experimental pain¹³² and migraine.^{36,55} Given our findings, gray matter variation in S1 may reflect persistent reorganization with pain.

Multiple mechanisms may explain structural differences in S1 under chronic pain. Animal studies show reorganization after injury, including in interneurons and glial cells.^{32,49,70,71} One possibility is that neuroplastic changes result in more efficient processing requiring reduced cortical area.⁴³ Chronic pain is also associated with reductions (eg, numbness) and alterations (eg, paresthesias) in somatosensory processing related to structural changes. Alternatively, S1's glial activation, implicated in nociceptive hypersensitivity,⁷⁰ could explain thickening as increased gray matter allocation for pain processing. Further, S1 thickening could reflect increased resolution of sensory signals, helping differentiate between chronic pain and acute, high-priority sensory information, such as Broca area thickening correlated with increased pitch discrimination.¹⁰²

Subcortical findings in persistent pain reveal increases in the right thalamus, parts of the basal ganglia, and decreases in the cerebellum. These changes, like cortical thickness alterations, are specific to the Persistent pain group, suggesting a delayed onset and partial recovery with pain resolution. The thalamus and basal ganglia are key regions for motivated behavior, with functional connectivity changes observed in patients with chronic pain.^{23,107,145} Previous studies have reported subcortical volume loss^{6,60,91,97} and gain,^{91,117,120,143} with inconsistencies possibly related to age³⁰ or pain type.²⁹ Larger volumes in these regions may reflect reorganization in neurons and glia, including stronger ascending pain pathways, central sensitization, or changes in neurotransmitter circuits,^{37,58,92} and the recruitment of endogenous pain modulatory mechanisms.³⁰ The cerebellum, involved in sensorimotor, emotional, reward, and executive functions, has shown both pain-related volume decreases^{17,85,118} and increases.^{29,119,124,134,143} These divergent findings may arise from its complex role in pain processing at multiple levels (sensory, emotional, and cognitive), as seen in migraine and depression studies.^{38,93,108}

Our findings expand on earlier reports by including a larger sample and more regions, with evidence of reversible changes in the occipital, parietal, and temporal cortex, cingulate, and precuneus. Effects on surface area were reversible for all regions, except for right S1, aligning with previous longitudinal studies showing gray matter change reversal upon pain resolution in areas including dlPFC, ACC, insula, vmPFC, and thalamus.^{61,98,113,114,121} Several cortical thickness regions align with the “short-term and persistent reversible” hypothesis, in which New and Persistent chronic pain groups differ from Controls but not from each other, suggesting that changes track pain but are unrelated to its duration.^{2,7} Alternatively, some regions, like M1 and bilateral insula, show no significant group differences even in this sample of nearly 35,052 individuals, in contrast to previous reports.^{17,29,86,142} Among subcortical volumes, caudate, putamen, and thalamus offer suggestive evidence in favor of irreversible changes.

Overall, conceptualization of pain effects on the brain continues to evolve, from damage, as earlier hypotheses suggested,⁶ to (mal) adaptive plasticity and resolution with pain recovery.⁹¹ Our findings show that most pain-related brain changes exhibit complete or partial reversal, although evidence for risk, irreversible, and reversible effects requires more nuanced research. Furthermore, while there are numerous possible cellular/molecular explanations for the structural changes we

observed in chronic pain, they remain speculative, and future work is needed to directly link the 2.

Our formulation does not imply that all forms of chronic pain are interchangeable. There are likely to be different mechanisms at play with different pain conditions,¹²⁶ subsets of pain conditions, such as musculoskeletal,⁸² and comorbidities.⁴¹ However, a growing body of work shows that there are common factors underlying multiple pain conditions, including shared genetic risk,^{47,148} hippocampal changes,⁴¹ psychosocial correlates,^{42,47} and medical sensitivities that are not localized to any particular body site.^{9,131} These findings suggest both common cross-condition and condition-specific mechanisms underlie chronic pain conditions.

5. Limitations

This study has several limitations. First, chronic pain categories were defined based on reported pain at baseline and MRI assessments. Without precise pain onset dates, we cannot confirm that the Persistent chronic pain group had pain between the 2 time points. Differentiating between predisposing and irreversible changes remains a challenge for large, longitudinal studies.^{13,62}

Second, to maximize sample size, the definition of Persistent chronic pain did not consider which body site(s) had pain. However, chronic pain conditions have a shared genetic factor,¹⁴⁸ musculoskeletal pain often spreads as it becomes chronic,^{27,131} and nociplastic pain can shift within a person.⁶⁶

Third, an important consideration for pain–brain associations is whether they are mediated by comorbid psychosocial conditions, such as depression and sleep.

Future studies are needed to understand the pathways by which pain may lead to brain alterations, such as changes in behavior and comorbid psychosocial conditions.

Fourth, the UK Biobank predominantly consists of White European individuals aged 40 to 69, limiting the applicability of our findings to broader populations.

6. Conclusions

This is the first large-scale study of gray matter effects as a function of new, persistent, and resolved chronic pain. We found widespread decreases in surface area and varying patterns of cortical thickness and subcortical volume in individuals with persistent chronic pain. These changes may contribute to the accrual of comorbidities observed in chronic pain conditions. Importantly, gray matter changes are largely reversible upon pain resolution, with the exception of S1 loss in surface area and hypertrophy in cortical thickness as well as thalamic and basal ganglia volumes. Overall, our findings highlight a role for brain plasticity in chronic pain that is more consistent with pain-related adaptations than with predispositional factors or pain-induced damage.

Conflict of interest statement

The authors have no conflict of interest to declare.

Acknowledgements

U.S. National Institute on Drug Abuse (NIDA) R01 (DA046064) Wager/Friedman.

Data used in this study are publicly available, registration required, <https://biobank.ndph.ox.ac.uk/showcase/>.

Supplemental digital content

Supplemental digital content associated with this article can be found online at <https://links.lww.com/PAIN/C581>, <https://links.lww.com/PAIN/C582>.

Article history:

Received 17 February 2026

Received in revised form 20 July 2026

Accepted 17 August 2026

Available online 2 October 2026

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